SPOT 1

1. Septal infarct (Anteroseptal infarct is also okay)
2. 94/min or close to 100/min
3. Decreased number of gap junction

SPOT 2

1. d and e
2. b
3. 'a' is the atrial pressure wave. It occurs when ventricles contract and push against a closed AV valve, this causes a rise in atrial pressure leading to 'a' wave.

SPOT 3

1. Cardiac contractile cell
2. Because of the prolonged refractory period
3. 'T' wave

SPOT 4

1. Baroreflex
2. Medulla
3. Less stretch occurs → decreased impulses to medulla → increased sympathetic stimulation (increases heart rate and contractility and increased blood vessel constriction that increase the TPR) and decrease parasympathetic activity (increases heart rate)

SPOT 5

1. Palpatory and Auscultatory
2. Korotkoff sounds; Produced due to turbulence in blood flow
3. MAP = Diastolic pressure + 1/3rd Pulse pressure

SPOT 6

1. A = SA node; B = AV node; C = Bundle of His; D = Purkinje fibers; E = Bundle branch; F = Internodal fibers
2. SA node → 100 – 110; AV node → 40 – 60; Purkinje fibers → 20 – 40
3. Inc. dromotropic effect – Sympathetic stimulation; Dec dromotropic effect – Parasym sti

SPOT 7

1. 1 → Isovolumetric contraction; 2 → Ventricular ejection; 3 → Isovolumetric relaxation; 4 → Rapid filling; 5 → Atrial contraction
2. 0.8 sec
3. Isovolumetric relaxation